

Muhammad Huzaifa *Software Engineer*

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📍 Karachi, Pakistan 🗣 Medium

Software Engineer with experience in AI/ML applications, full-stack web development, and automation. Built award-winning AI systems (3rd Prize, Solana Breakout Hackathon) and deployed production-ready LLM applications (RAG chatbots, semantic recommenders, AI weather summaries). Passionate about research at the intersection of machine learning, automation, and scalable systems.

📁 PROFESSIONAL EXPERIENCE

Rehani Soko LLC — Remote  Apr 2023 – present | United States
Software Developer & Technical Support Specialist

Nov 2024 – Present (Full-Time)

Promoted from contract to full-time role based on performance and ownership of core systems.

- Engineering automated pipelines to collect and clean **50k+ property listings/month**, improving dataset reliability by **35%**
- Leading development of features for a **Vue.js + Django** real estate platform, enhancing data visualization and search
- Building robust data preprocessing and consistency-check scripts to support downstream **AI/ML models**

Apr 2023 – Oct 2024 (Contract / Part-Time)

- Developed and maintained data ingestion pipelines for global real estate listings
- Supported frontend feature development and internal tooling during early platform scaling

Lovum Group — Remote  Jan 2024 – Dec 2025 | United States
Software Engineer (Part-Time / Side Engagement)

- Building **Next.js + Python** applications i.e. **EduEssential** and **Waridi**, focusing on high-performance, scalable UI and backend systems.
- Contributed to **API design & integration**, ensuring seamless data flow across frontend, backend, and ML-driven features.

Algorithm Consulting (Pvt) Ltd  May 2024 – Oct 2024 | Karachi, Pakistan
Junior Developer

- Contributed to the development of an **ASP.NET**  **enterprise application**, implementing new features and resolving high-priority bugs.
- Worked closely with senior engineers to **optimize backend performance** and improve release stability.

CrossWing Inc. — Remote  Mar 2023 – Nov 2023 | Ontario, Canada
Full Stack Developer

- Built and maintained **React.js web applications** used in AI-powered robotics solutions.
- Collaborated with **AI/ML engineers and product managers** to deliver features bridging robotics hardware and web-based interfaces.

Software Engineer Grade A1

- Automated **data extraction workflows** using Import.io [↗](#) , improving collection efficiency for client datasets.

EDUCATION

NED University of Engineering & Technology [↗](#)

2019 – 2023 | Karachi, Pakistan

Bachelor of Engineering

- Major in Computer and Information Systems Engineering.
- Cumulative GPA: 3.289/4.0
- Relevant Coursework: Data Structures and Algorithms, Software Engineering, Artificial Intelligence, Machine Learning.

PUBLICATIONS

An Integrated Approach to Robotic Object Grasping and Manipulation

2023

Huzaifa, M., et al. (2023). *An Integrated Approach to Robotic Object Grasping and Manipulation using 6D Pose Estimation*. arXiv:2411.13205

arXiv [↗](#) PDF [↗](#)

Smart Shelf Advertising using Real-Time Product Segmentation and Interaction on Low-Cost Edge Devices

2025

Huzaifa, M., et al. (2025). Smart Shelf Advertising using Real-Time Product Segmentation.

DOI:10.13140/RG.2.2.34475.96800.

ResearchGate [↗](#)

SKILLS

Frontend Development — Vue.js • React.js • Next.js • Tailwind CSS

Backend Development — Node.js • Express.js • Django • Flask • ASP.NET [↗](#) (C#)

AI/ML — OpenAI API • LangChain • Embeddings & Vector Search • Scikit-learn • Pandas • NumPy

Data Mining & Automation — Selenium • BeautifulSoup • Puppeteer • Python Scripting

Databases — PostgreSQL • MySQL • MongoDB • Firestore • GraphQL

PROJECTS

Optimal Robotic Arm Gripper, FYP [↗](#)

2023

Robotics, Computer Vision, AI

- The Final Year Design Project aims to improve object grasping and manipulation in warehouses.
- Designed an autonomous robotic arm using dual monocular cameras for **6D object pose estimation** and **ML-driven grasp strategies**.

- Implemented **kinematic & inverse kinematic modeling** for real-time object retrieval in warehouse scenarios.

F1GPT — Formula One AI Chatbot [↗](#)

Aug 2025

Next.js 15, RAG, GPT-5, LangChain, Astra DB, Puppeteer

- Built a **Retrieval-Augmented Generation (RAG) chatbot** for F1 insights, driver stats, and rules.
- Developed scrapers & chunking pipelines, stored embeddings in **Astra DB**, and integrated a **GPT-5 API layer** for contextual Q&A.

LLM Semantic Book Recommender [↗](#)

Jul 2025

LangChain, Vector DB, GPT APIs

- Created an **embeddings-based recommender** that indexes books into a vector database for semantic retrieval.
- Used **LLM re-ranking** to provide **context-aware, beyond-keyword recommendations** with natural-language justifications.

AI Weather App [↗](#)

2023

Next.js 13, OpenAI API, Tremor Charts, Tailwind CSS

- Developed a weather dashboard fetching forecasts from **Open-Meteo API** and generating **LLM-written daily summaries**.
- Visualized time-series data with Tremor charts, styled with Tailwind.

AWARDS

3rd Prize — AI Track (awarded \$15,000) [↗](#)

Solana Breakout Hackathon (Global)

Selected among **8,300+ global submissions** for building *Agent Arc* [↗](#), a non-custodial AI-driven trading terminal on Solana. Led frontend development using **Next.js 15**, Privy authentication, wallet integration, and AI/ML API execution.